



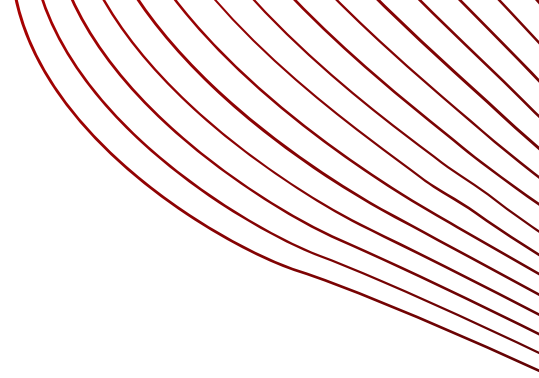
DEPARTMENT OF PHYSICS, IIT DELHI

DIRECT CLASSIFICATION OF PHASE OBJECTS AND PHASE RECONSTRUCTION USING DEEP LEARNING

**BY
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TAMAL MAJUMDER**

UNDER THE SUPERVISION OF
PROF. KEDAR B. KHARE

MAY 9TH, 2025

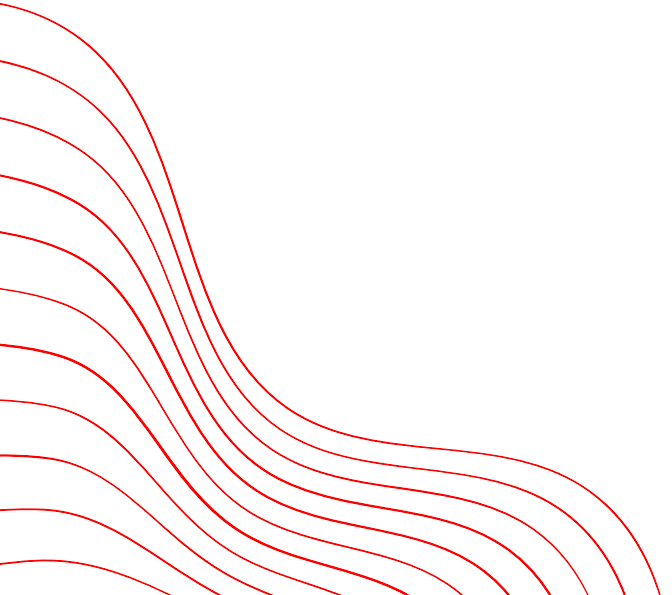


OUTLINE

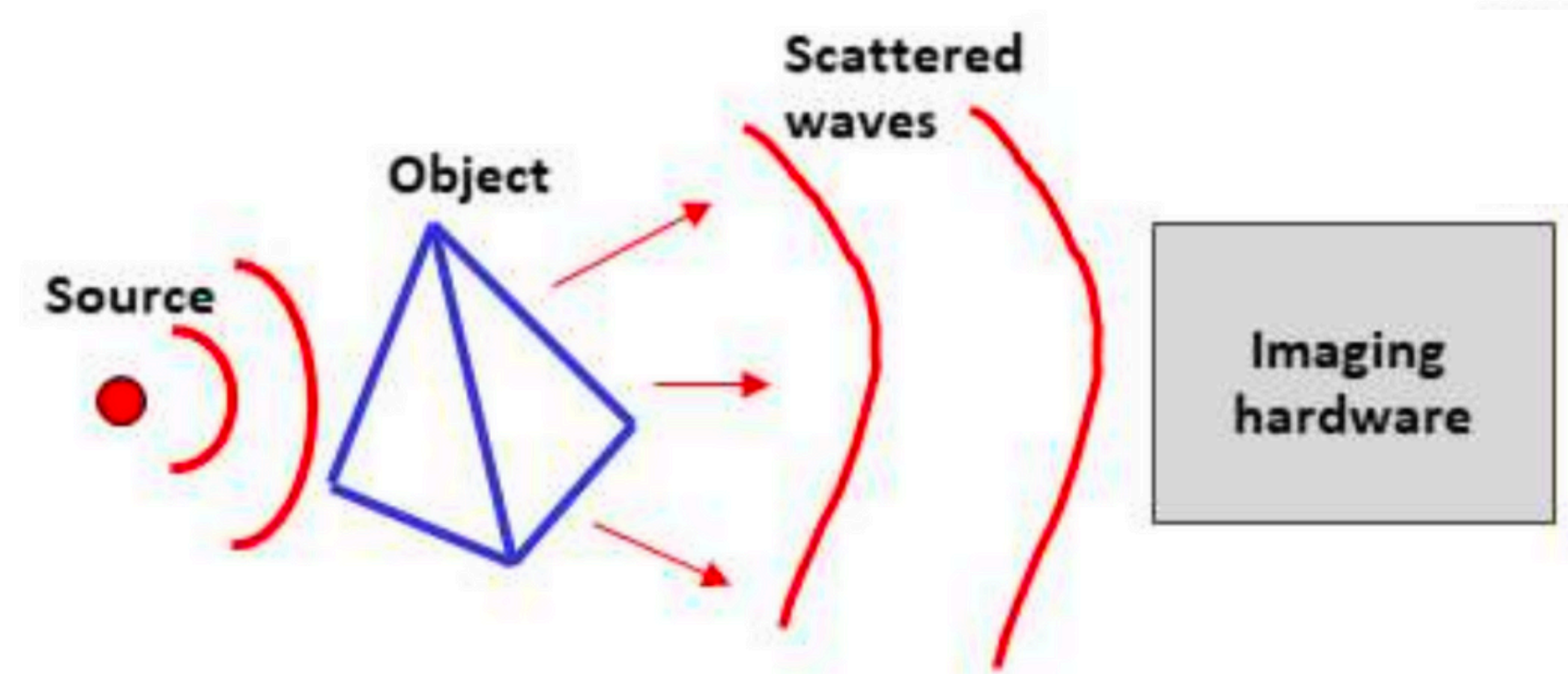
1. UNDERSTAND THE PROBLEM

2. CLASSIFICATION

3. RECONSTRUCTION



WHY PHASE IS IMPORTANT IN IMAGING



Ref: Fourier optics and computational imaging by k. khare , M Butola & S Rajora

PHASE CONTAINS THE STRUCTURAL INFORMATION OF THE IMAGE.

ENHANCES CONTRAST FOR TRANSPARANT SAMPLES

Part 1

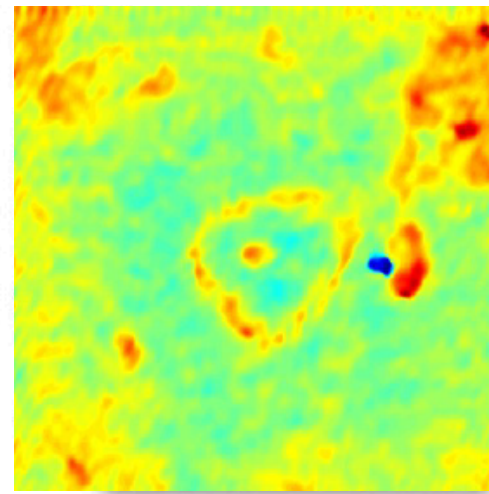
DIRECT CLASSIFICATION OF HOLOGRAMS

PROBLEM STATEMENT

TRADITIONAL APPROACH



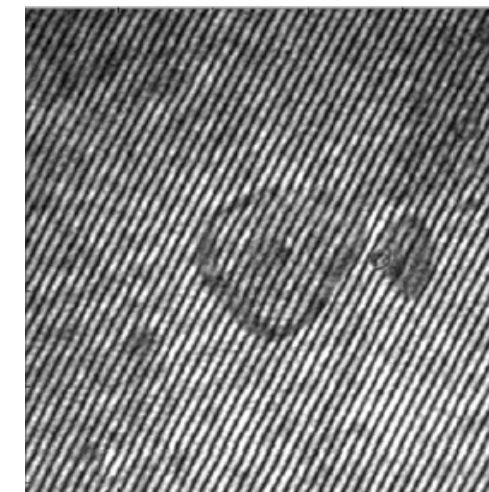
Hologram



Reconstruction

CLASSIFICATION

OUR APPROACH



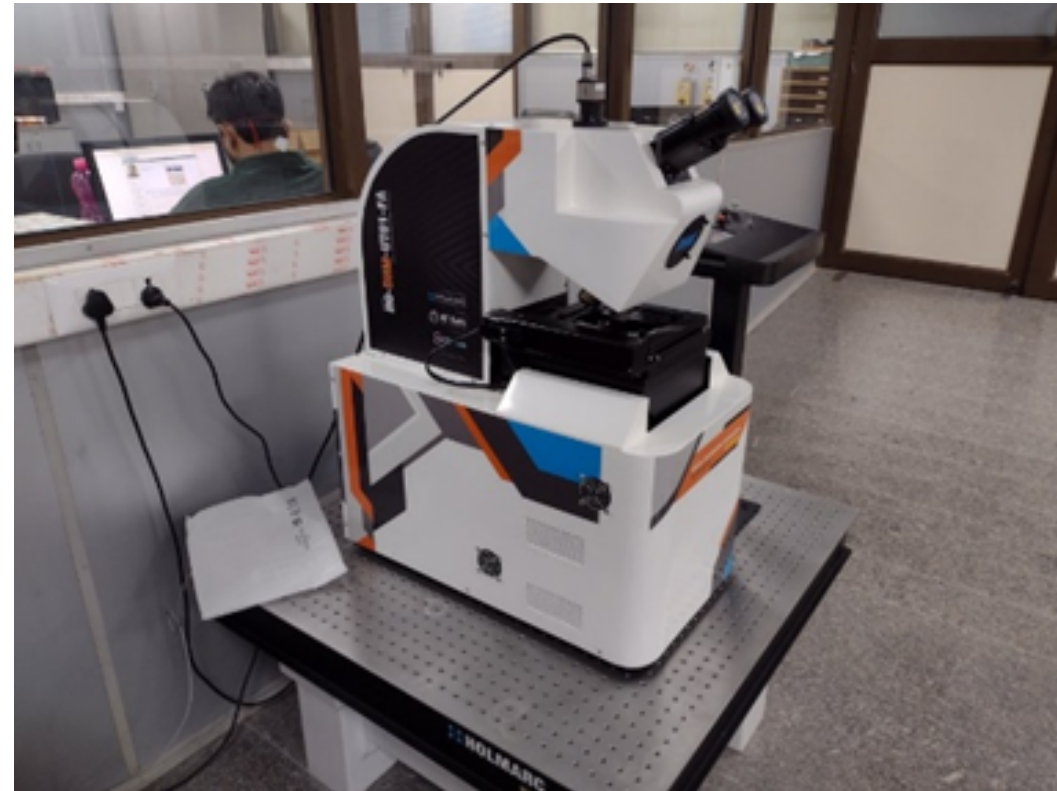
Hologram



CLASSIFICATION

SO OUR APPROACH IS TO **CLASSIFY THE HOLOGRAMS DIRECTLY**
"BYPASSING THE RECONSTRUCTION STEP"

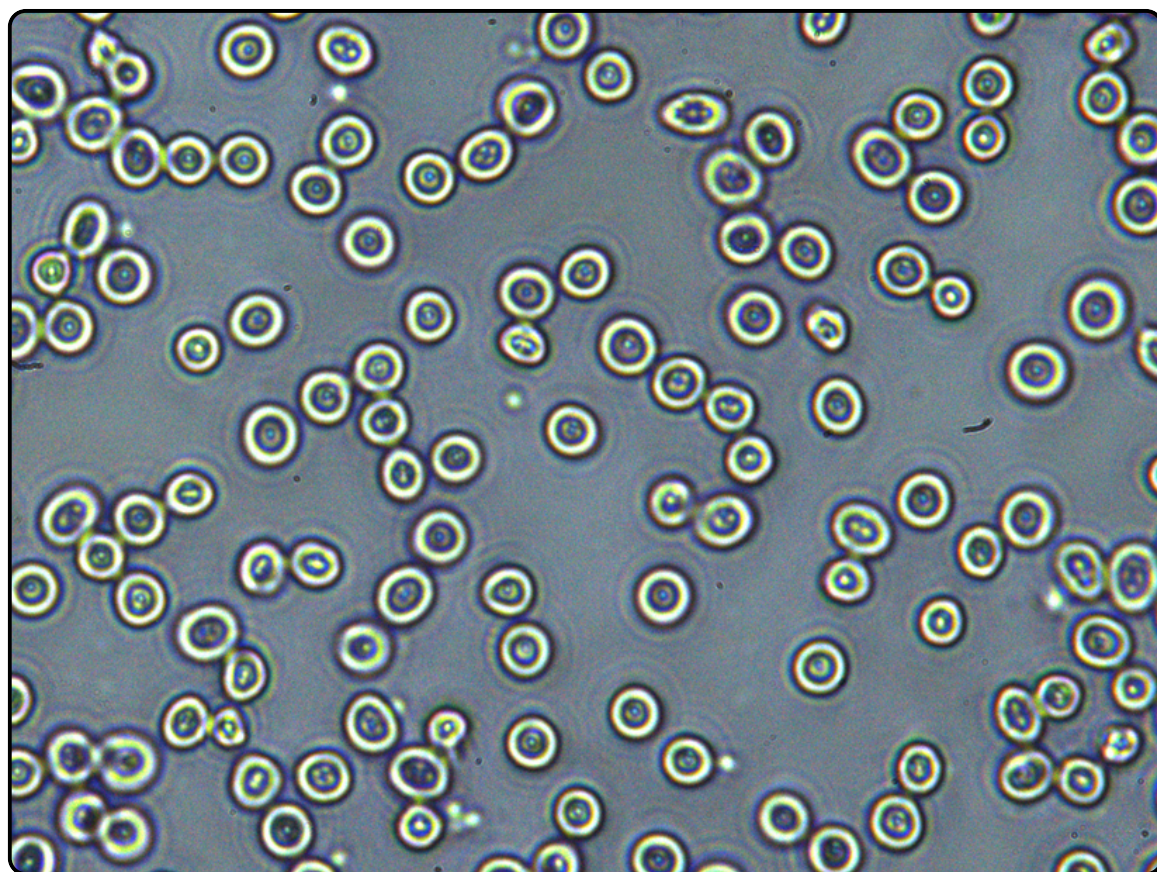
DHM MICROSCOPE:



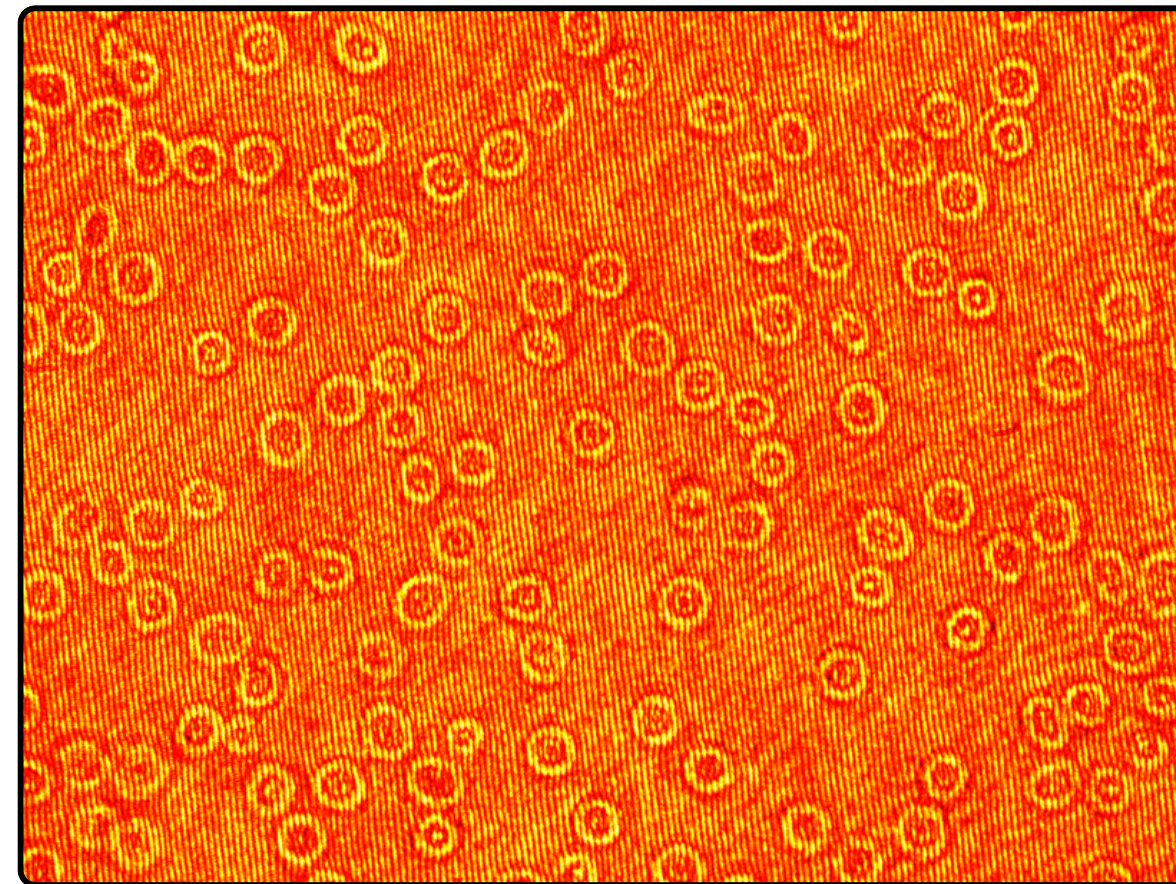
- Single Shot Acquisition
- Enhanced Resolution
- Unique iterative Optimization
- Medical & Biological applications
- Affordability & accessibility

Components

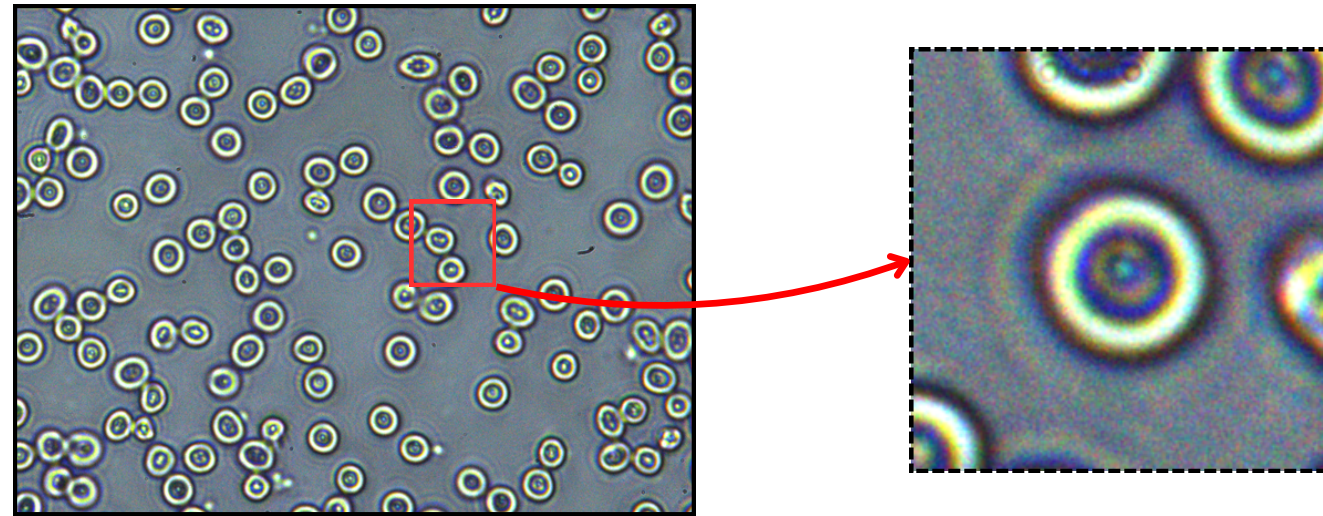
- **Illumination Source:** Coherent monochromatic red light LASER beam (650nm).
- **Specimen stage:** motorized stage allowing specimen for fine adjustments across multiple axes.
- **Interferometer:** Off-axis holography - Mach-Zehnder interferometer.
- **Digitizing Camera:** CCD or CMOS sensor - pixel size $3.5\ \mu\text{m}$ - captures holograms and bright field image at 3056×1542 dimension simultaneously.
- **Reconstruction Algorithms:** Iterative optimization techniques, including sparsity-based methods.



BRIGHTFIELD IMAGES

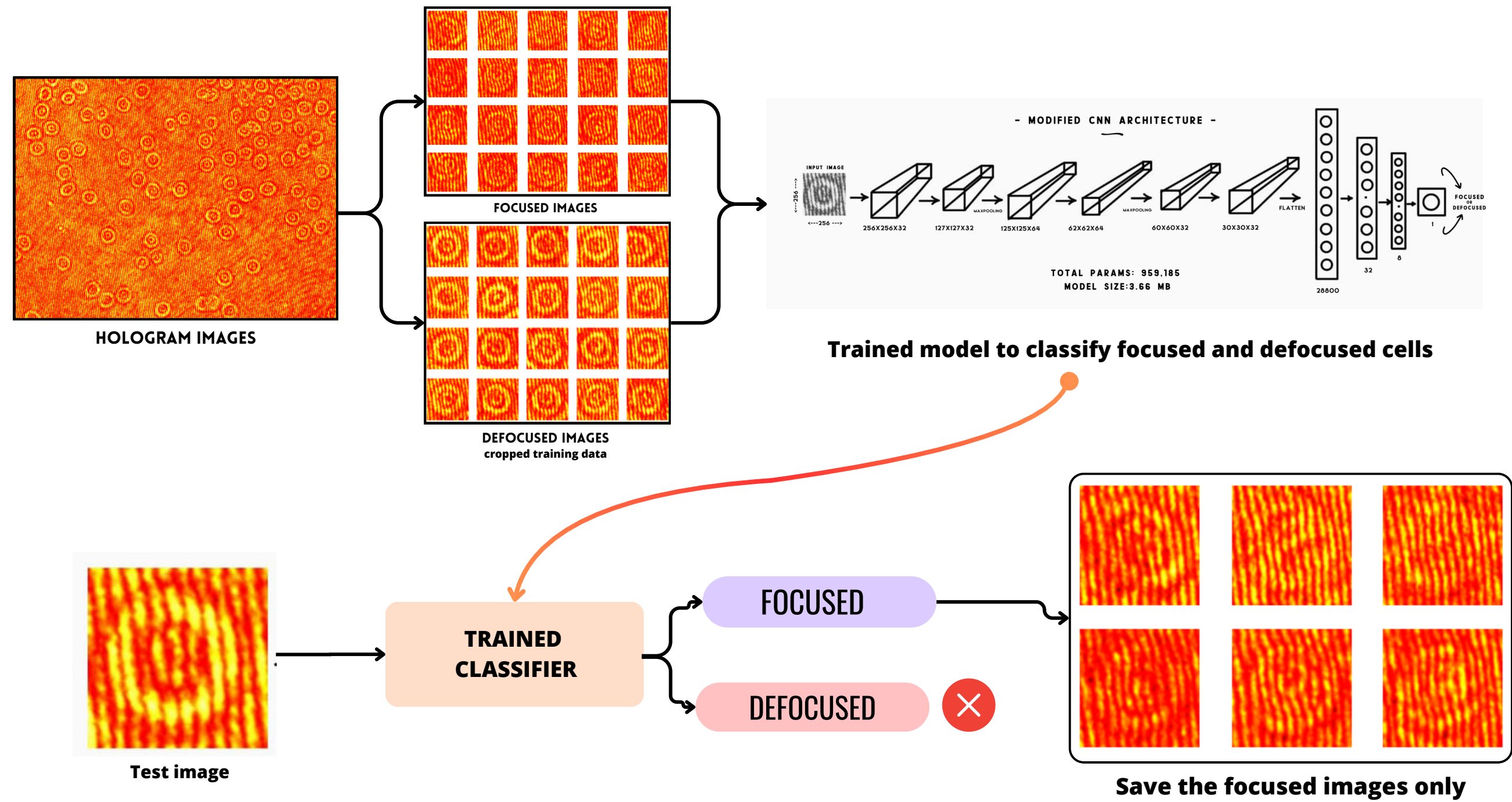


HOLOGRAM IMAGES



- **MANUAL CROPPING:** We've to choose ROI of the cell we interested for further processing. → **AUTO SEGMENTATION**
- **Don't know whether the selected cell is focused or defocused.** → **CLASSIFICATION**
- **ITERATIVE ALGORITHM:** takes a lot of time even for small ROIs. → **DEEP LEARNING-BASED PHASE RECONSTRUCTION.**

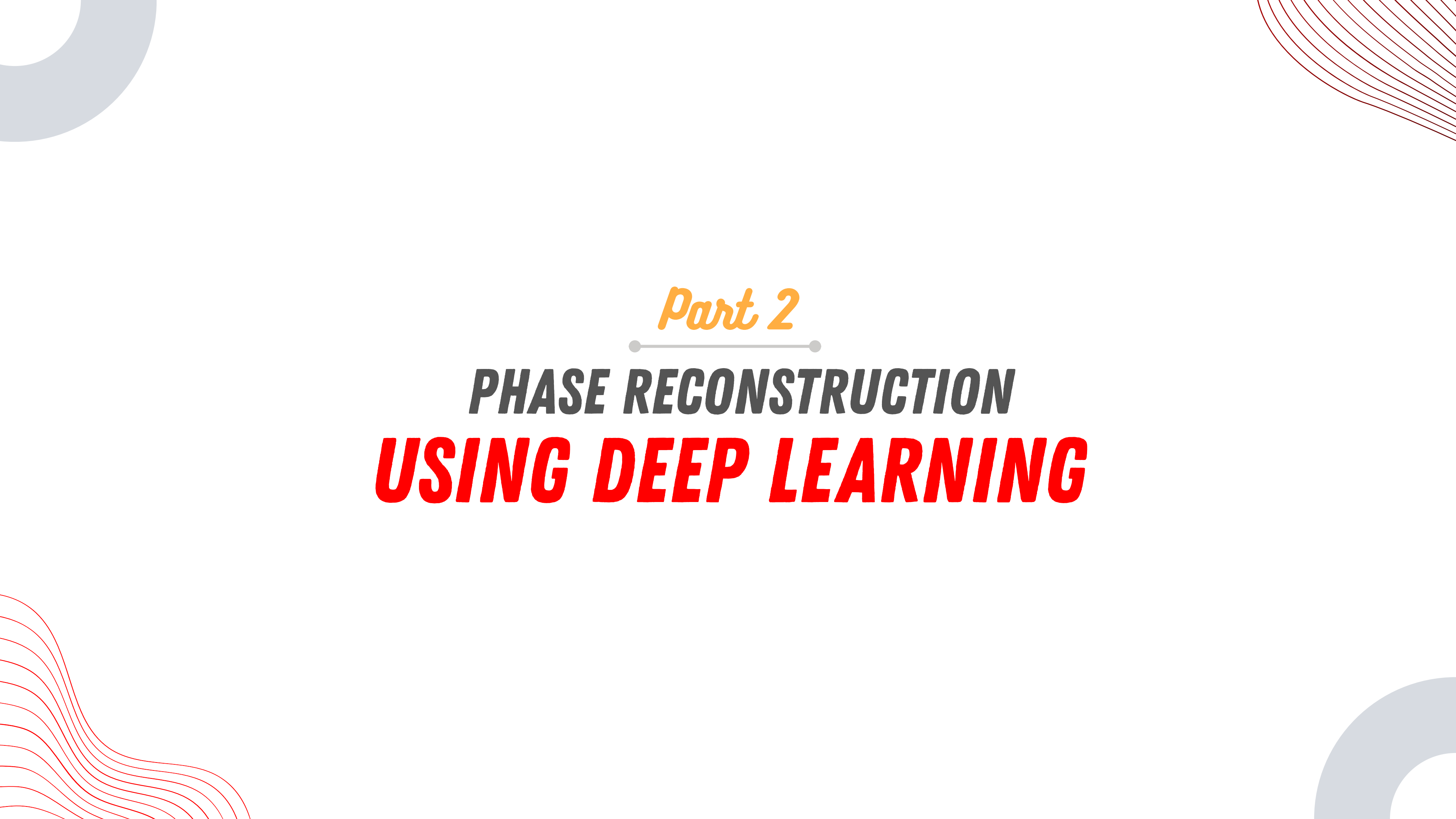
CLASSIFICATION SCHEME:



CLASSIFICATION SUMMARY:

Model	Total Params	Total Size (MB)	Accuracy
CNN	59,207,169	224.98	0.9938
CNN 56 MB	14,842,041	56.62	0.9938
CNN 28 MB	7,430,753	28.35	0.9938
CNN 14 MB	3,743,265	14.28	0.9877
CNN 3 MB	959,185	3.66	0.9938
CNN 1.8 MB	498,169	1.90	0.9815
AlexNet	15,669,313	59.77	0.9815
AlexNet Simplified	8,590,401	32.77	0.9691
ANN Basic	8,392,897	32.02	0.9259
PCA to ANN (99%)	127,489	0.4863	0.9877
PCA to ANN (50%)	10,113	0.0386	0.9867

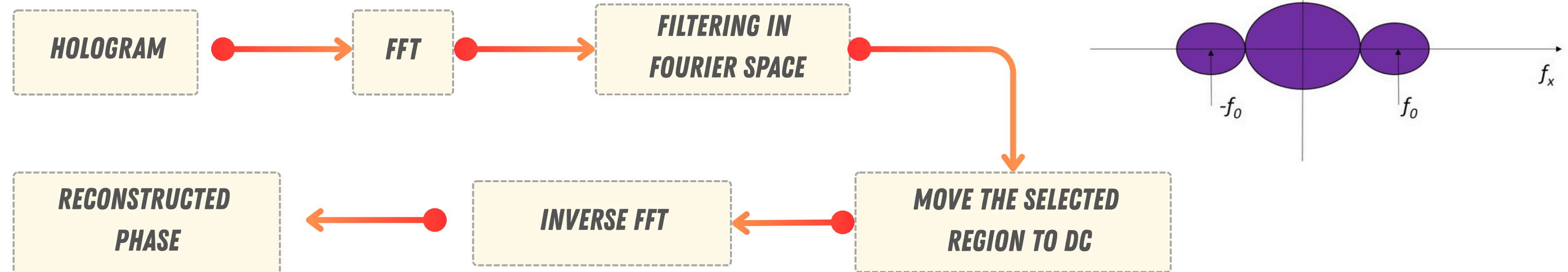
Table 3.3: Final Merged Model Comparison Table



Part 2

***PHASE RECONSTRUCTION
USING DEEP LEARNING***

FOURIER TRANSFORM METHOD:



PHASE SHIFTING METHOD:

$$\begin{aligned}
 H_1 &= |R|^2 + |O|^2 + 2|R||O| \cos(\phi_O - \phi_R), \\
 H_2 &= |R|^2 + |O|^2 - 2|R||O| \sin(\phi_O - \phi_R), \\
 H_3 &= |R|^2 + |O|^2 - 2|R||O| \cos(\phi_O - \phi_R), \\
 H_4 &= |R|^2 + |O|^2 + 2|R||O| \sin(\phi_O - \phi_R).
 \end{aligned}$$

4 SHOT HOLOGRAPHY **

$$\phi_O(x, y) = \phi_R(x, y) + \arctan \left(\frac{H_4 - H_2}{H_1 - H_3} \right)$$

OPTIMIZATION TECHNIQUE:

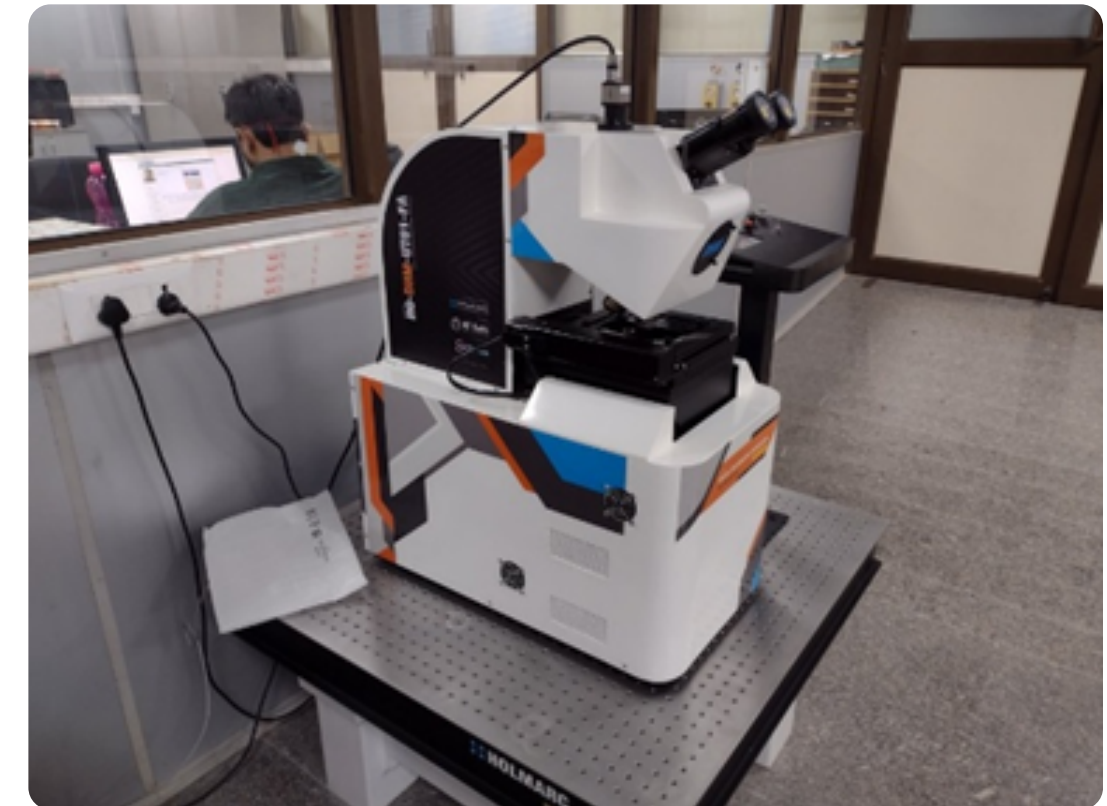
$$H(x, y) = |O(x, y)|^2 + |R|^2 + 2|R||O(x, y)| \cos(\phi_r(x, y) - \phi_o(x, y)).$$

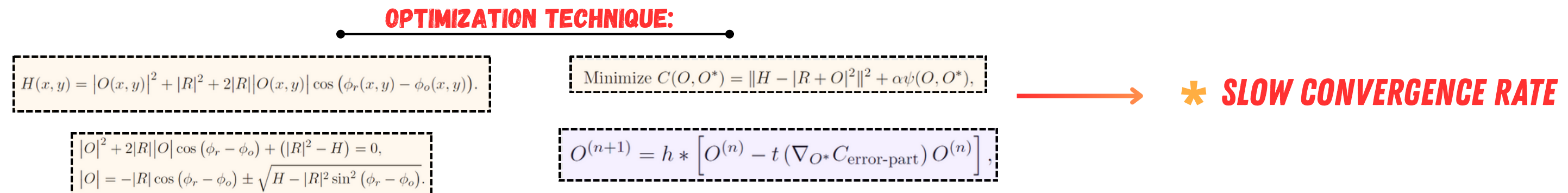
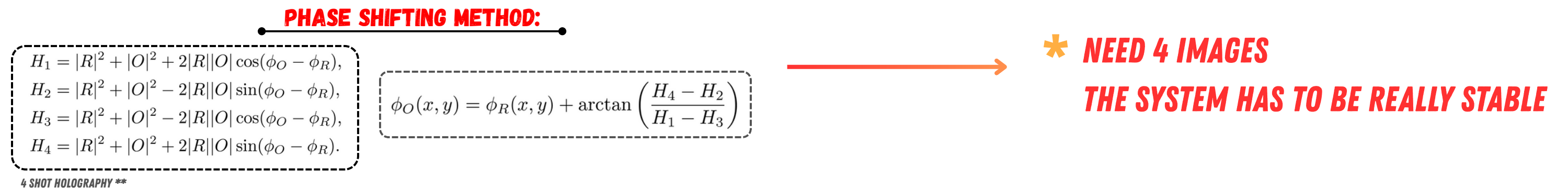
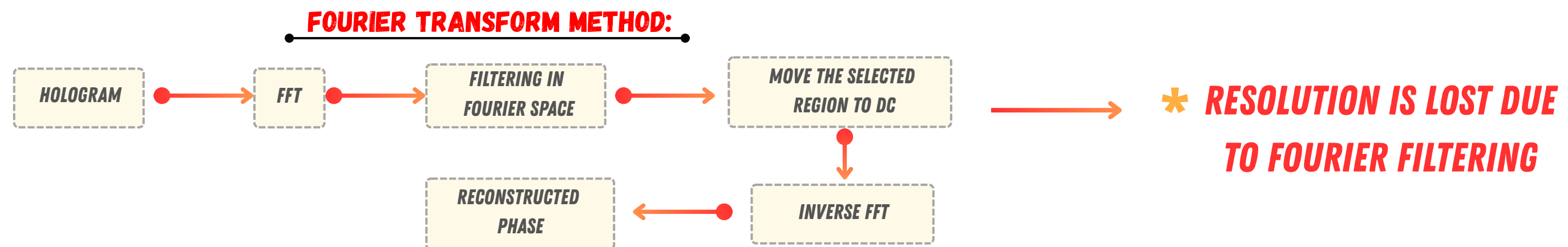
$$\begin{aligned} |O|^2 + 2|R||O| \cos(\phi_r - \phi_o) + (|R|^2 - H) &= 0, \\ |O| &= -|R| \cos(\phi_r - \phi_o) \pm \sqrt{H - |R|^2 \sin^2(\phi_r - \phi_o)}. \end{aligned}$$

$$\text{Minimize } C(O, O^*) = \|H - |R + O|^2\|^2 + \alpha\psi(O, O^*),$$

$$O^{(n+1)} = h * \left[O^{(n)} - t (\nabla_{O^*} C_{\text{error-part}}) O^{(n)} \right],$$

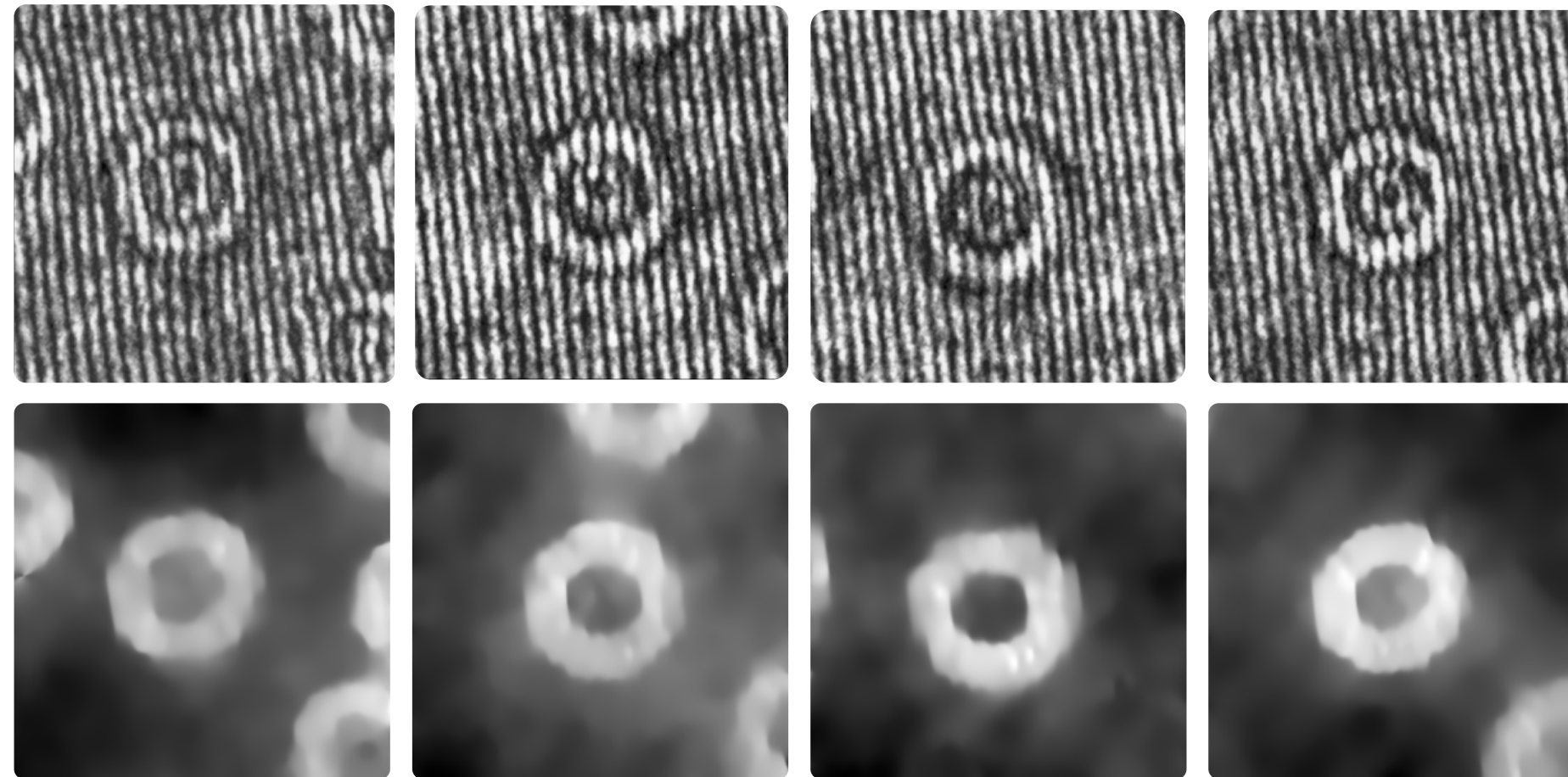
CONVERGES THROUGH ITERATIONS
LIVE CELL IMAGING



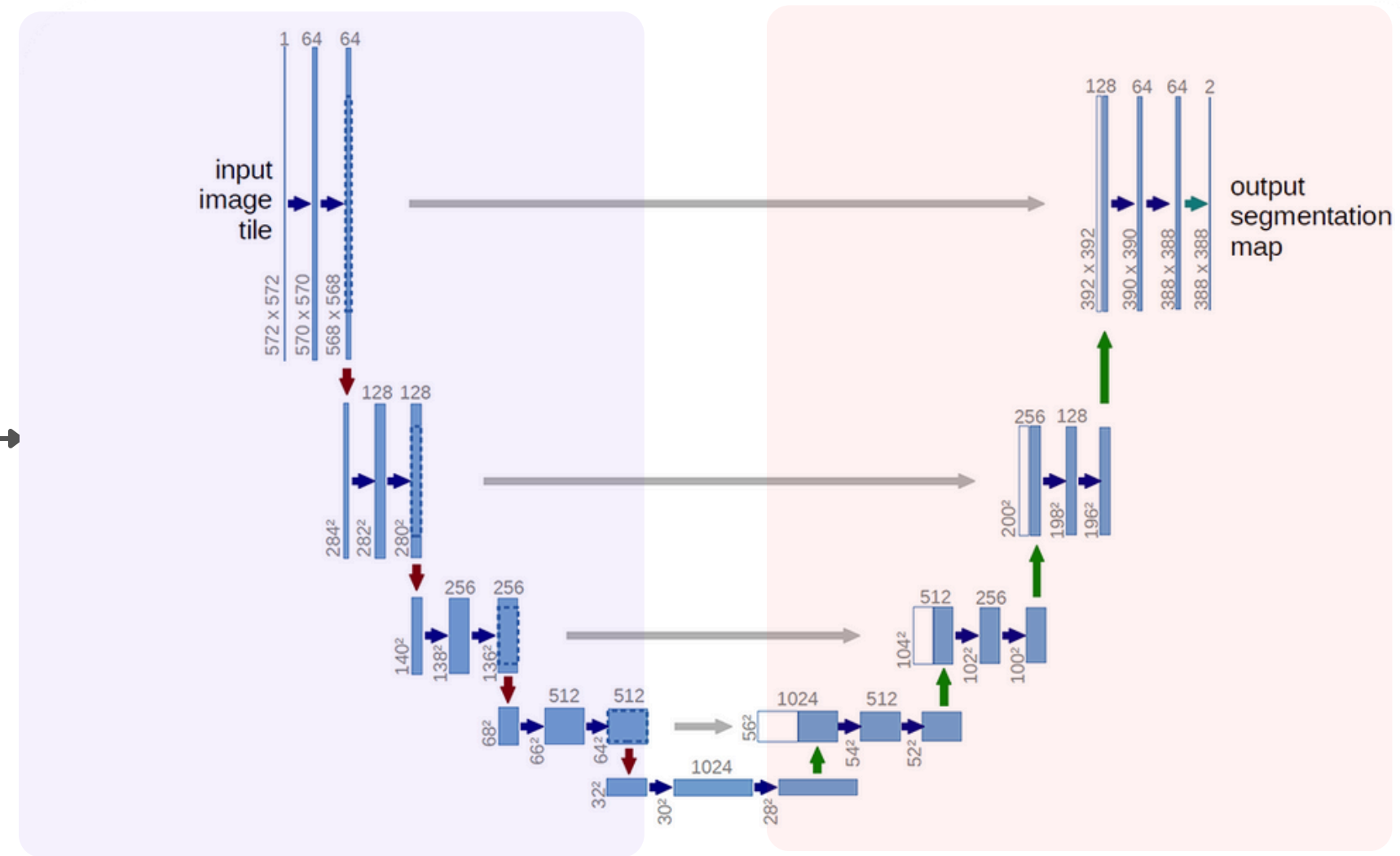
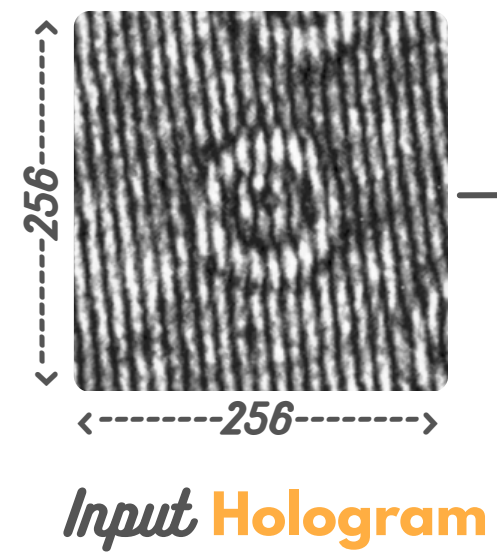


OUR GOAL IS TO BYPASS THESE LIMMITATIONS USING DEEP LEARNING.

DATASET DESCRIPTION



***WE HAD A DATASET OF 750 SUCH
HOLOGRAM AND RECONSTRUCTED PHASE PAIRS***

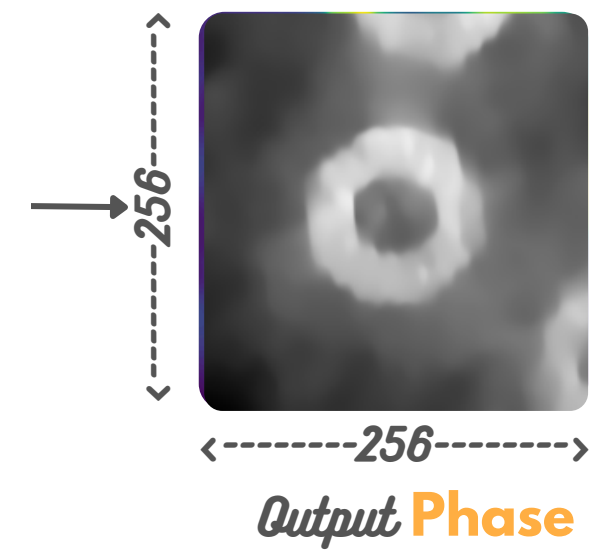


ENCODER

Encodes the most important pixels / information

DECODER

Decodes the encoded information back to original size



PHYSICAL CONSTRAINT INFORMED **LOSS FUNCTION**

$$\mathcal{L}_{\text{combined}} = \alpha \cdot \mathcal{L}_{\text{MSE}} + \beta \cdot (1 - \mathcal{L}_{\text{SSIM}}) + \gamma \cdot D_{\text{KL}}(P \parallel Q)$$

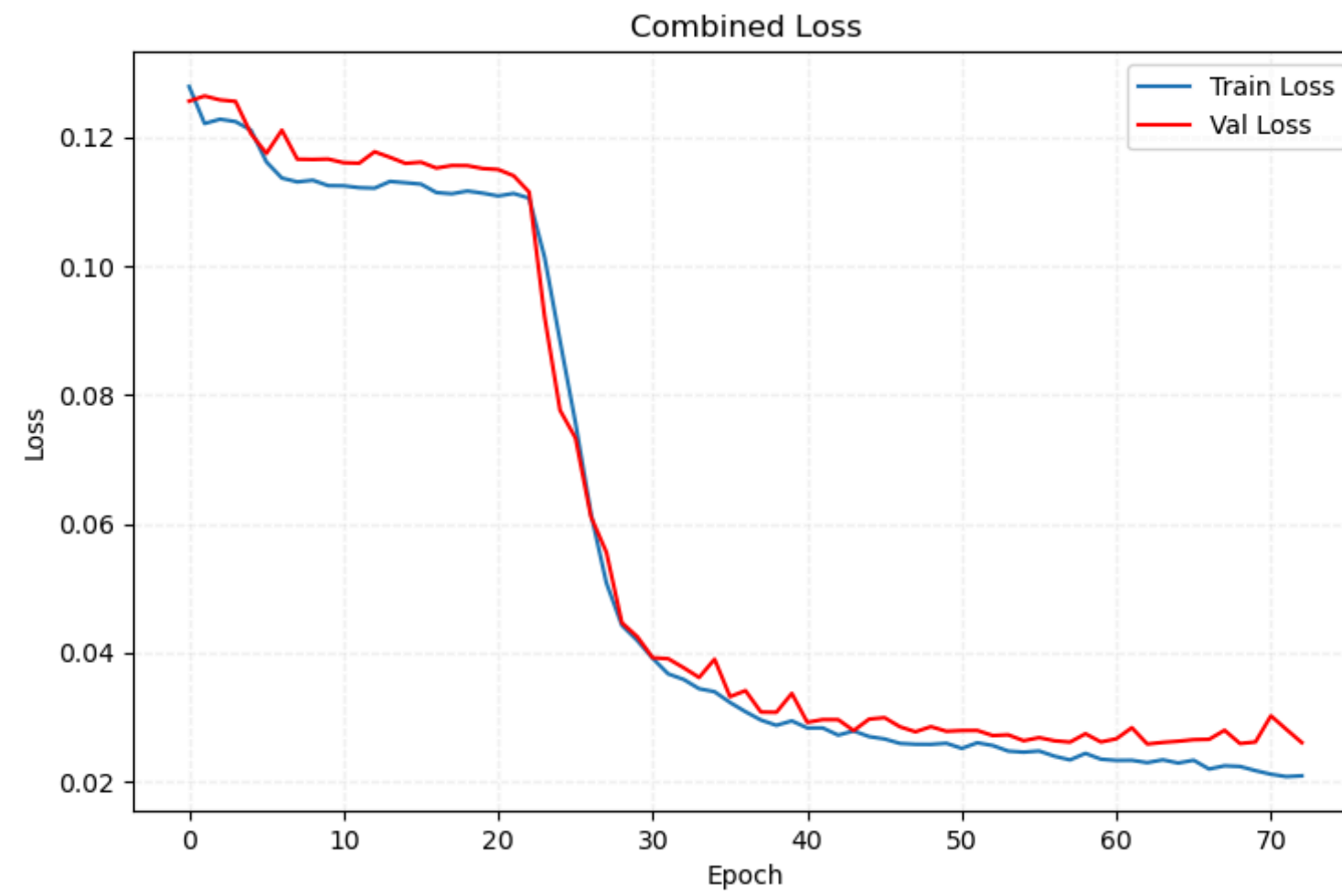
$$\underbrace{\frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2}_{\text{(MSE)}}$$

$$\underbrace{\left(1 - \frac{(2\mu_y\mu_{\hat{y}} + C_1)(2\sigma_{y\hat{y}} + C_2)}{(\mu_y^2 + \mu_{\hat{y}}^2 + C_1)(\sigma_y^2 + \sigma_{\hat{y}}^2 + C_2)}\right)}_{\text{(SSIM)}}$$

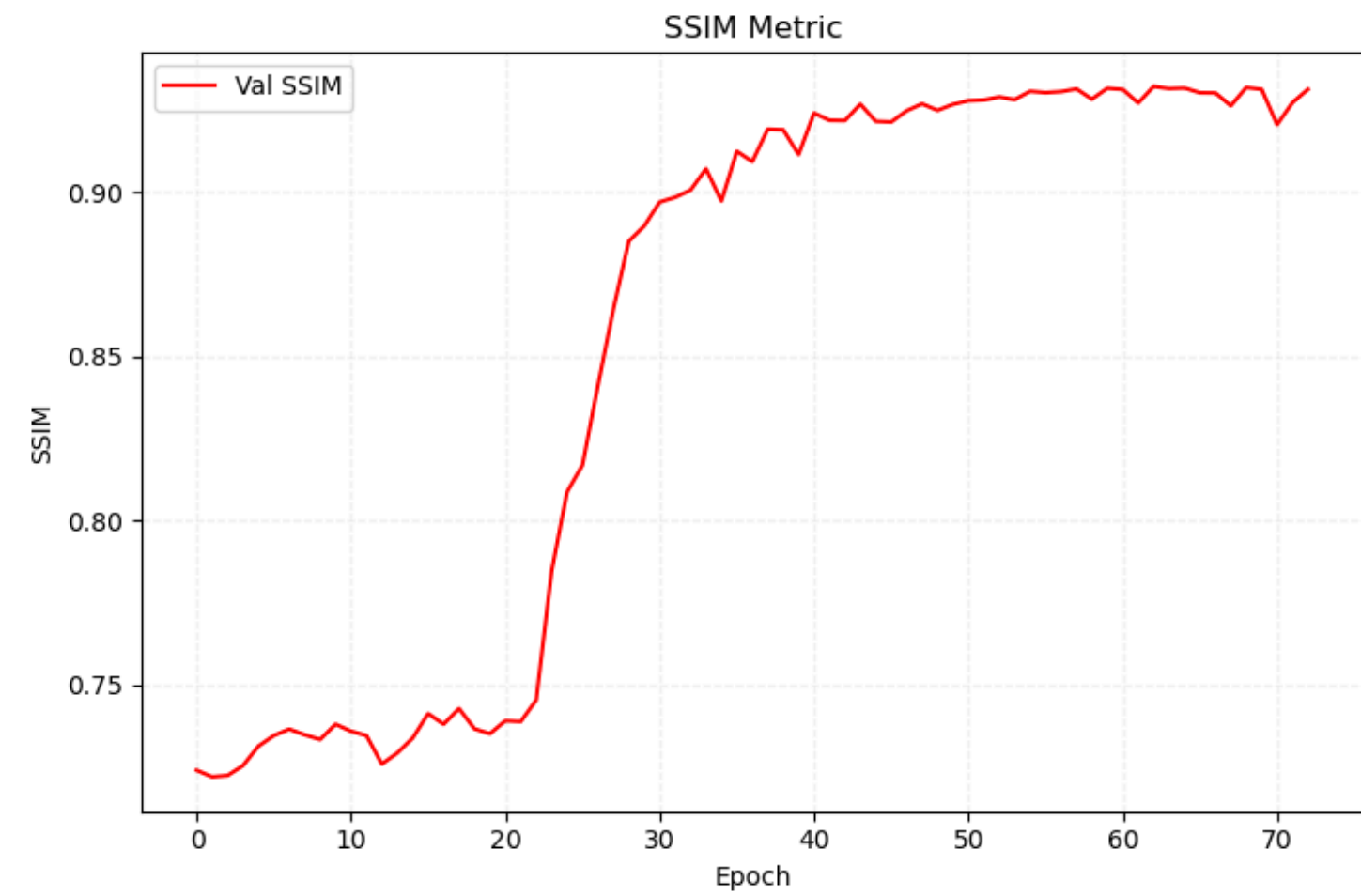
$$\underbrace{\sum_x P(x) \log \frac{P(x)}{Q(x)}}_{\text{(KL Divergence)}}$$

PARAMETER TRACKING

LOSS



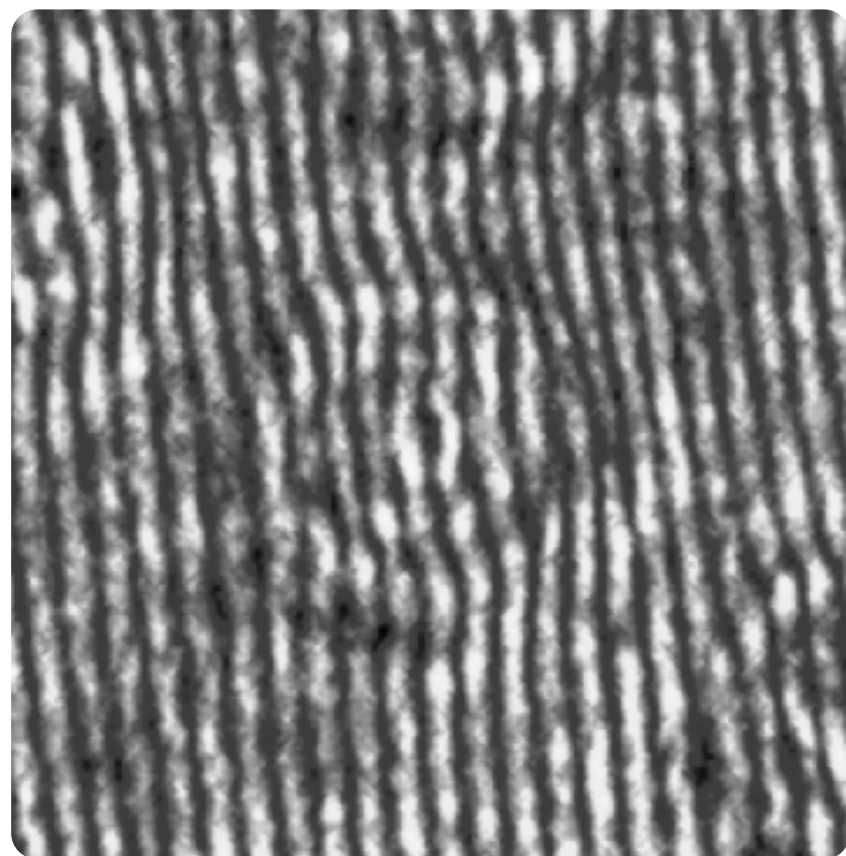
SSIM



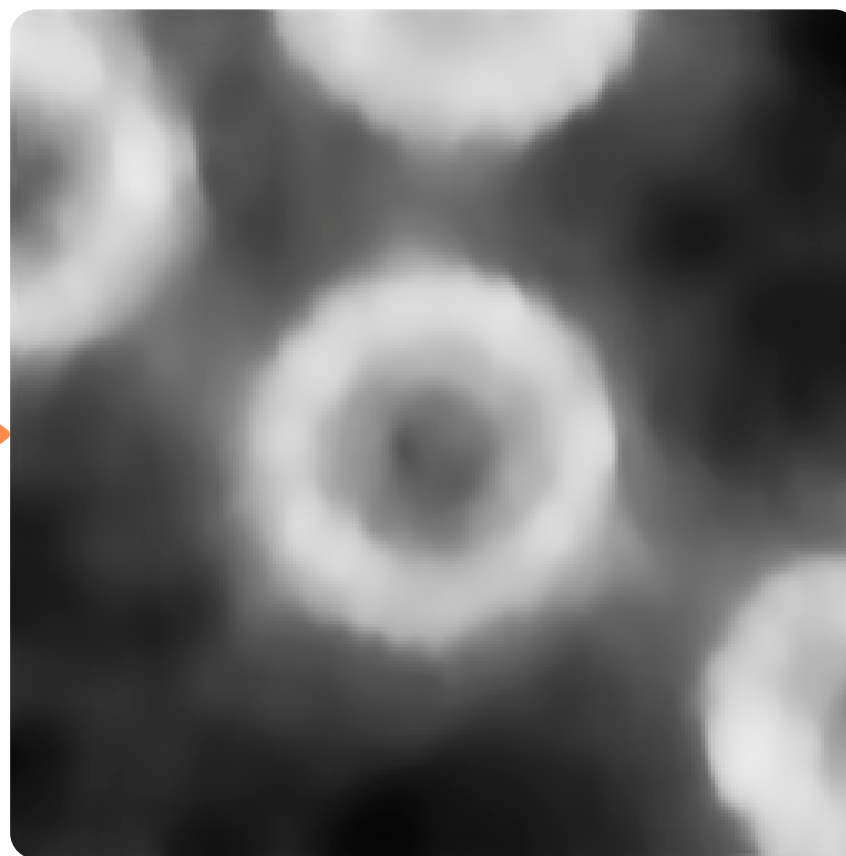


RESULTS & IMPLICATIONS

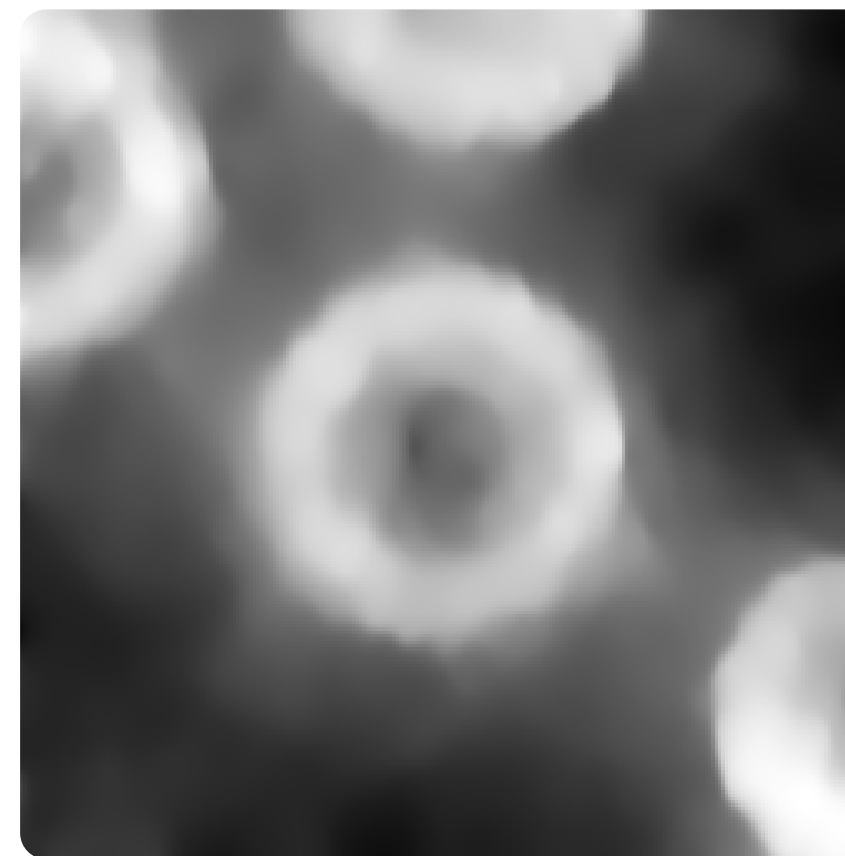
RESULTS



Input **Hologram**



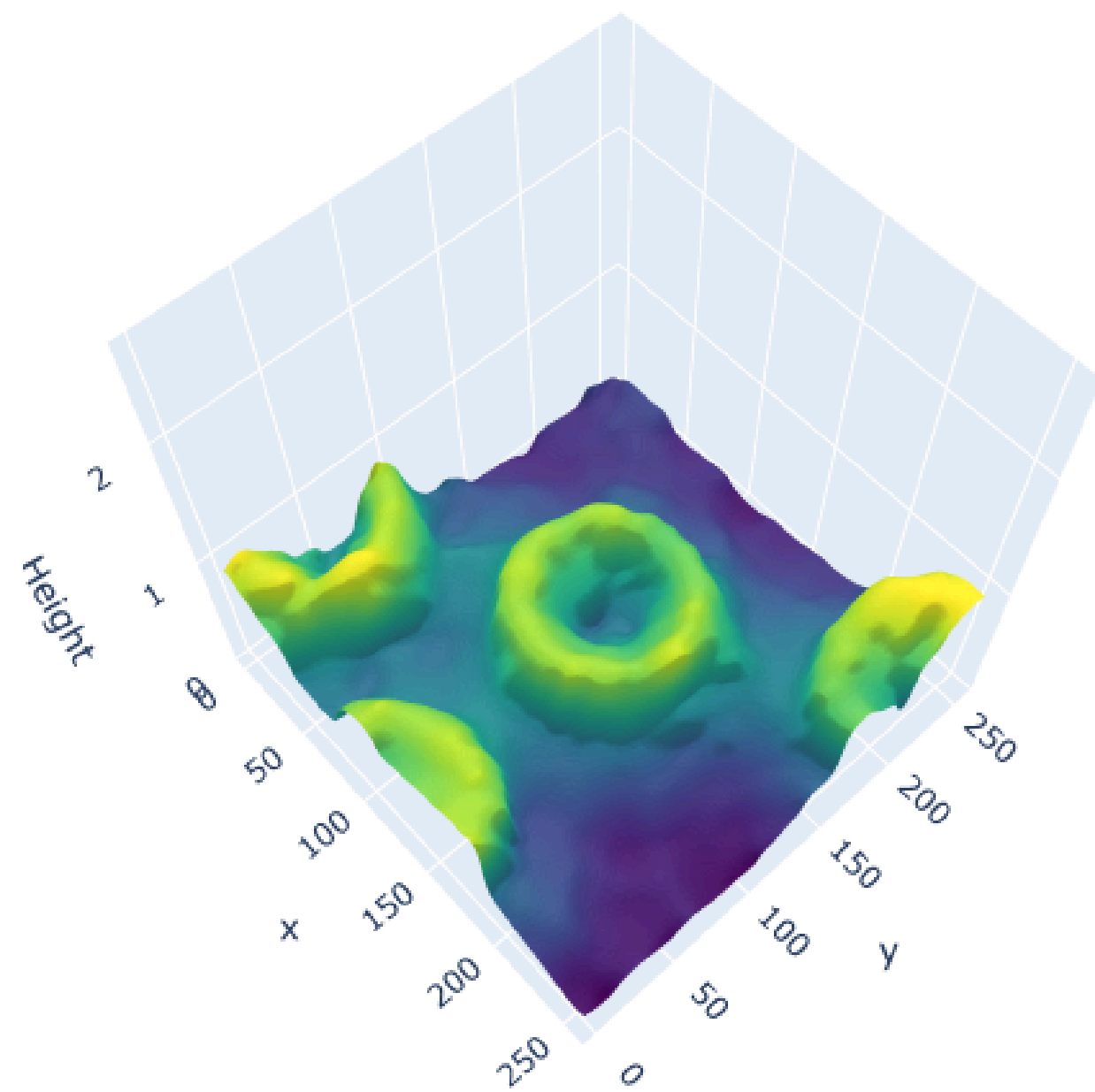
Predicted **Phase**



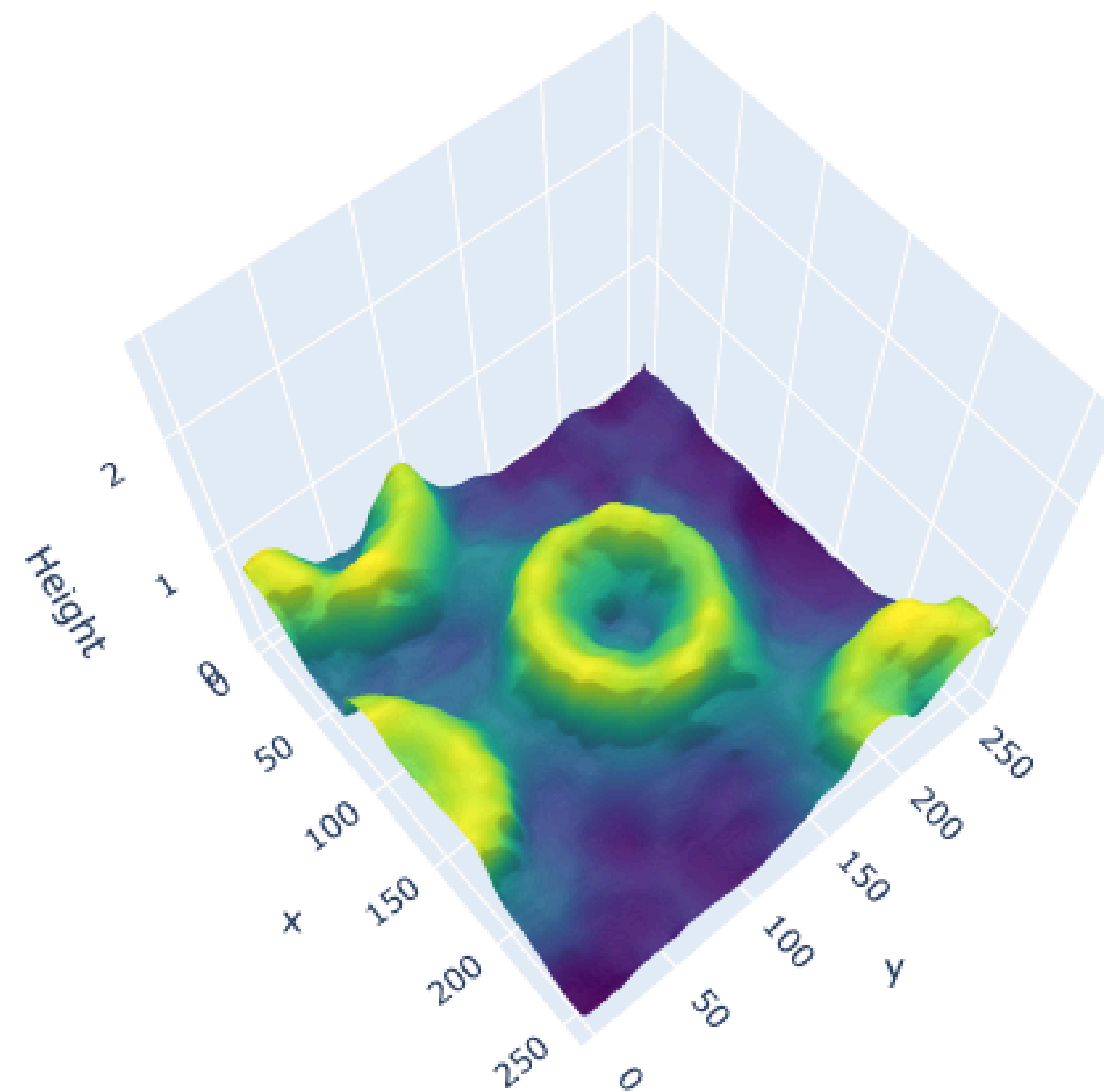
Actual **Phase**

SSIM : 0.9578

PHASE MAP IN 3D

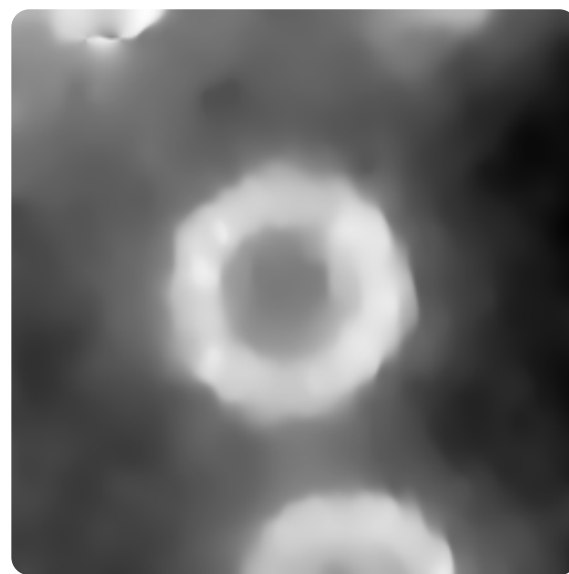


Actual



Predicted

SSIM : 0.9255

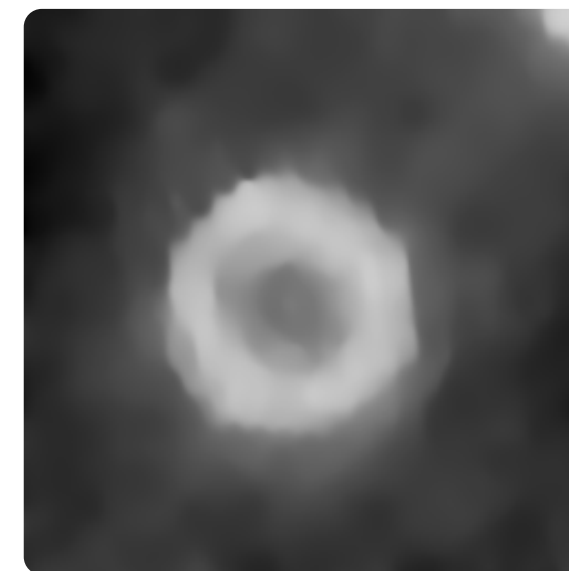


actual



pred

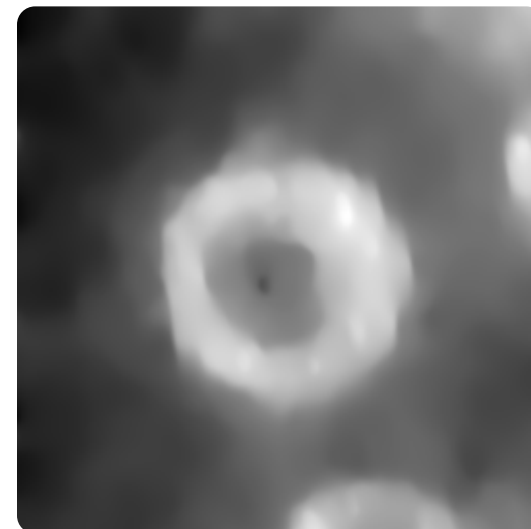
SSIM : 0.9543



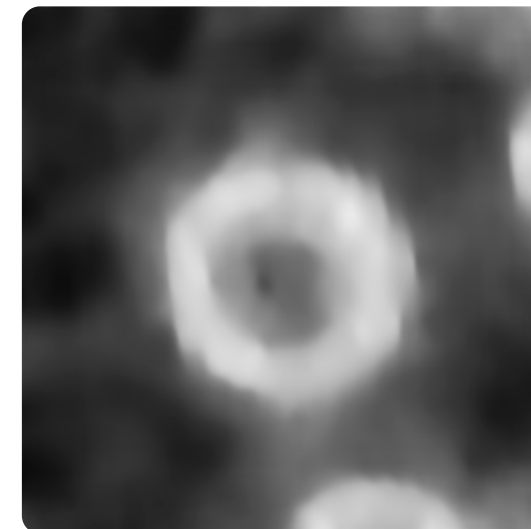
actual



pred



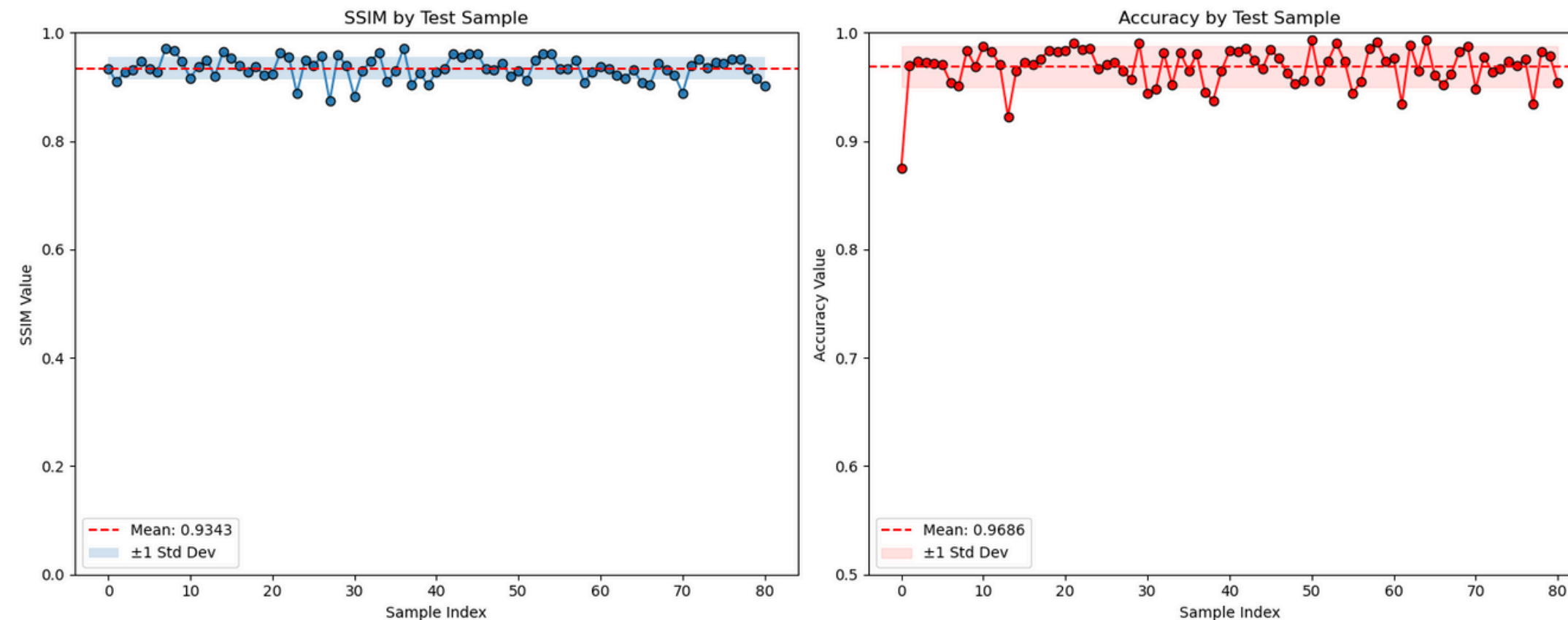
actual



pred

SSIM : 0.9324

STATISTICAL ANALYSIS OVER TEST DATA



The average SSIM we got was **93.43%** and the average accuracy was **96.86%**. The standard deviation for SSIM was **0.020** and for the reconstruction accuracy was **0.018**. for the test set. We chose SSIM to be the performance quantifier of our model. So, we can conclude that our model is able to reconstruct the phase with $(93.43 \pm 2)\%$ similarity. So the model's performance is quite reliable and stable.

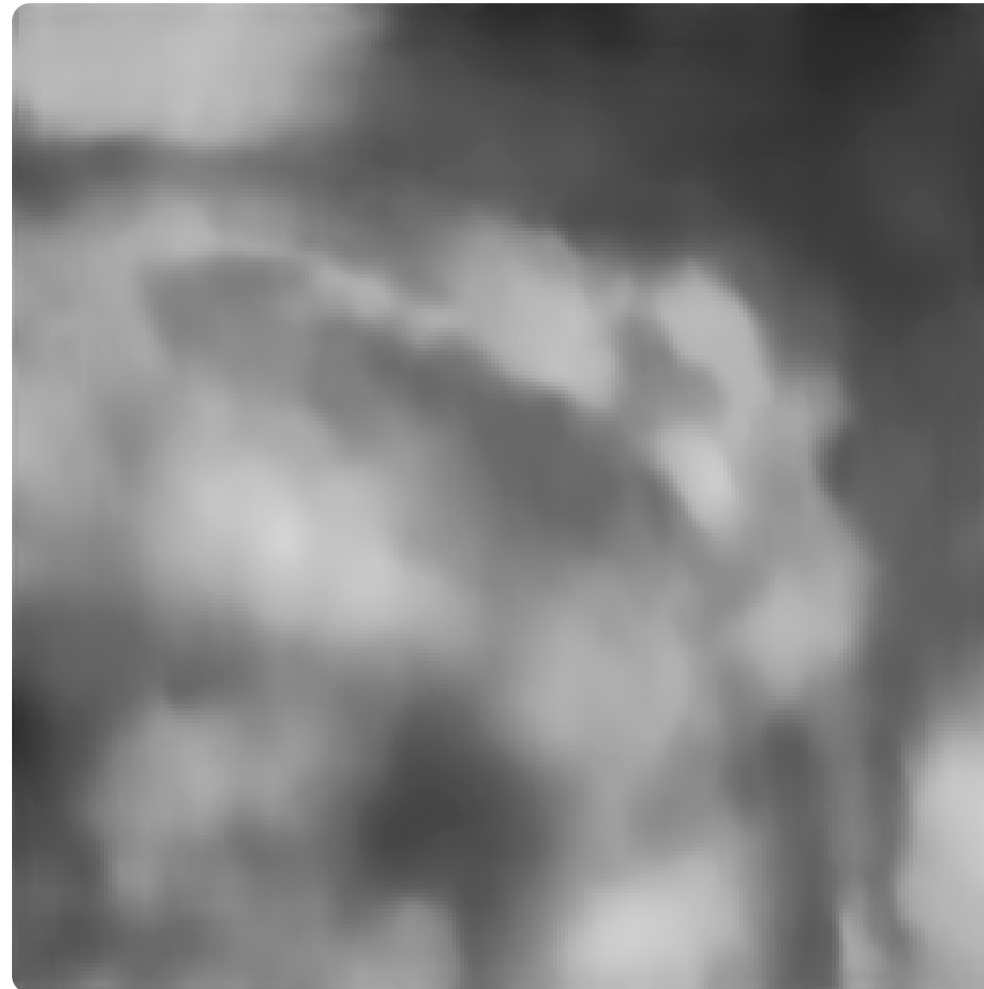
PERFORMANCE SCORE : $(93.43 \pm 2)\%$

MODEL PERFORMANCE ON NOVEL DATA

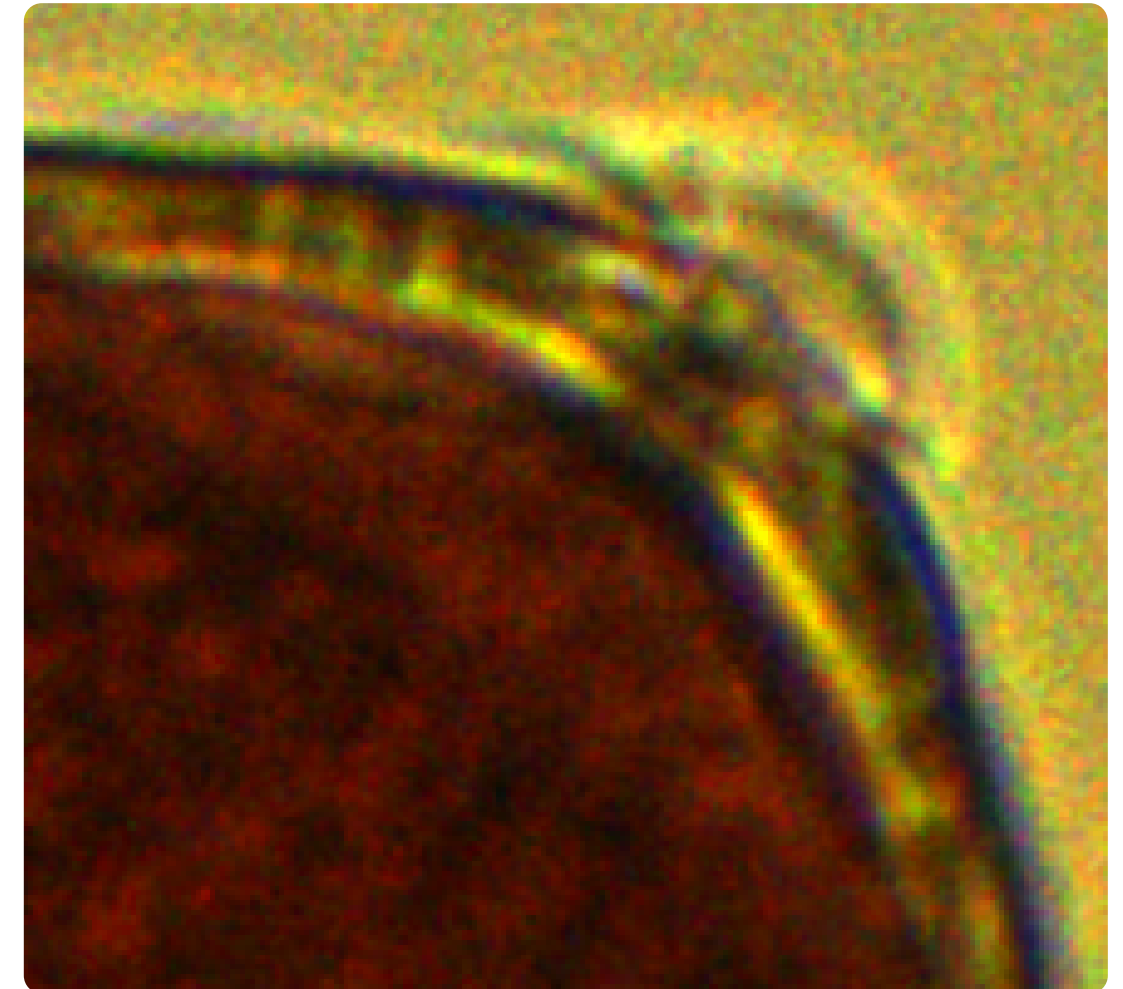
Input **Hologram**



Output **Phase**



Brightfield **Image**



Inference on **ANOTHER CLASS OF DATA**

HARDWARE USED

HAD ACCESS TO *NVIDIA RTX 3060 GPU*

**EFFICIENT IN DOING *PARALLEL COMPUTATION*
3584 CUDA CORES**

*Using the SOTA **iterative Algorithms***

10 SECONDS *ON AN AVERAGE**

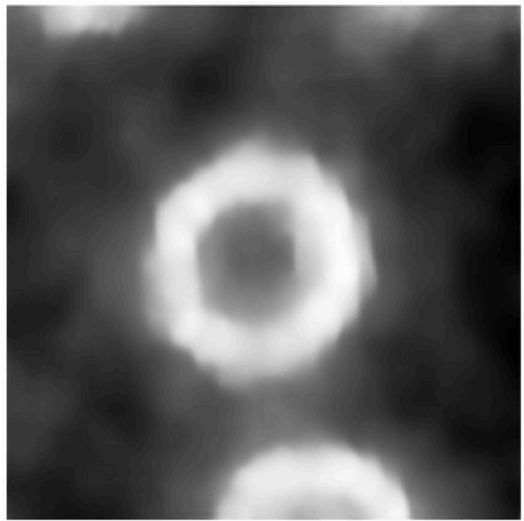
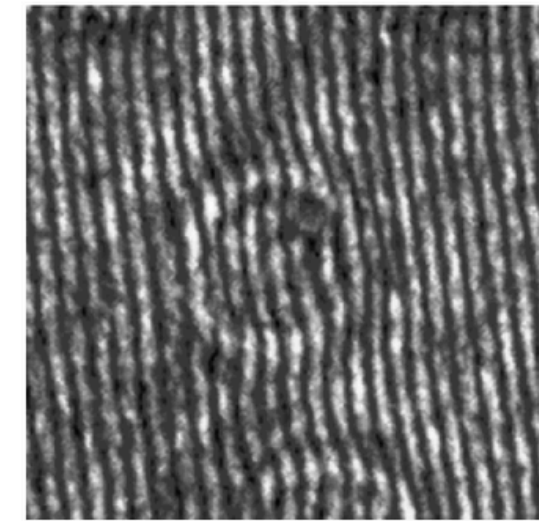
INFERENCE TIME

```
In [11]: 1 image_path = r'C:\Users\987ta\Desktop\FYP\Final year data and code stuff\test and result\holos\holos502.png'
2 save_path = r'C:\Users\987ta\Desktop\FYP\Final year data and code stuff\test and result\holos\Pred'
3
4 reconstruct_and_save(image_path, save_path, model, device='cuda' if torch.cuda.is_available() else 'cpu')
5
```

COMPUTE ON GPU

Original

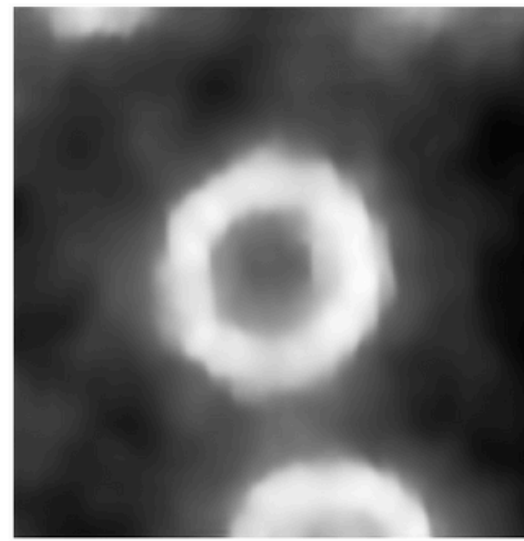
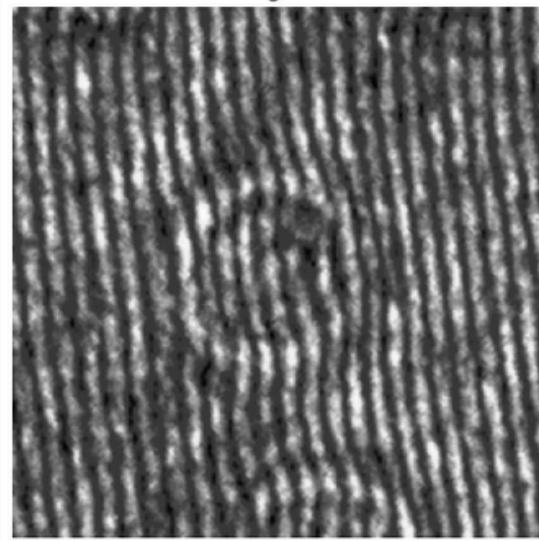
Reconstructed



Inference Time: 0.0020 seconds 2 MS

Original

Reconstructed



Inference Time: 0.0767 seconds 77 MS

COMPUTE ON CPU

COMPARISON TABLE

Approach	Hardware Used	Reconstruction Time	Improvement
Iterative Method	CPU	10 seconds	1×
Deep Learning Method	GPU (RTX 3060)	2 milliseconds	5000×
Deep Learning Method	CPU (AMD Ryzen 7 with 8 cores)	100 milliseconds	100×

REALTIME RECONSTRUCTION

ECONOMICAL PERSPECTIVE

15 Minutes of Compute time was needed

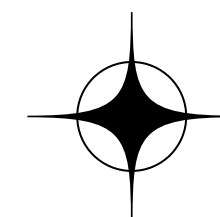
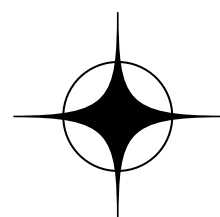
RTX 3060
90Wh

22.5 *Wh*

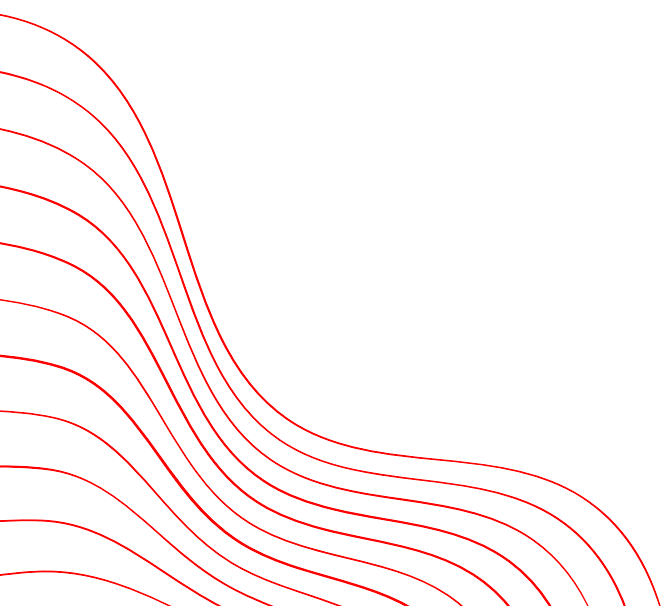
**CPU, RAM, DISPLAY
AND SEVERAL OTHER COMPONENTS**

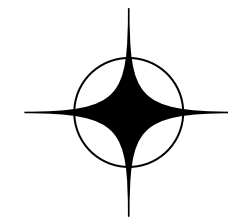
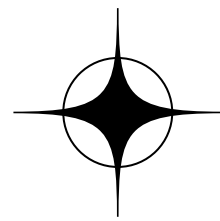
35 *Wh*

COST OF COMPUTING → **0.22 Rs**



QUESTIONS





THANK YOU

